

Sean Luka Girgis

Senior Cloud Capacity & Efficiency Engineer | AWS/GCP Capacity Planning | Python/SQL Forecasting

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Professional Summary

Senior Cloud Capacity & Efficiency Engineer with 20+ years of enterprise IT experience across capacity planning, infrastructure telemetry, data engineering, performance engineering, and cloud-scale analytics. Strong fit for cloud capacity and efficiency work through hands-on Python, SQL, PySpark, AWS, forecasting, telemetry pipelines, utilization analysis, rightsizing recommendations, and stakeholder-facing capacity reporting. Built forecasting and attribution-style data workflows using infrastructure telemetry from thousands of endpoints, supporting demand planning, bottleneck prediction, consolidation opportunities, cost-aware infrastructure decisions, and executive technology planning. AWS hands-on experience with transferable GCP cloud capacity concepts.

Professional Experience

Senior Capacity Planning & Cloud Data Engineer at CITI (2017-11 – 2025-12)

- Owned capacity-planning analytics for large-scale enterprise infrastructure by ingesting P95 telemetry from 6,000+ endpoints into automated Python/Pandas and SQL workflows for demand planning, risk detection, and executive reporting.
- Developed forecasting models using Prophet and scikit-learn to predict CPU, memory, and infrastructure capacity bottlenecks up to 6 months ahead with 90%+ accuracy, improving provisioning lead time and reducing reactive capacity decisions.
- Built AWS-based capacity analytics pipelines using S3 landing zones, AWS Glue ETL, Amazon Redshift workloads, and historical telemetry models to support cloud-scale utilization, trend, and forecasting analysis.
- Migrated heavy capacity forecasting workloads from on-prem Oracle processing to Amazon Redshift, enabling more scalable parallel analysis for infrastructure demand, utilization trends, and long-range planning.
- Identified underutilized infrastructure through data mining across thousands of endpoints, supporting rightsizing, hardware consolidation, spend reduction opportunities, and improved allocation of compute capacity.
- Produced capacity and efficiency reporting for senior technology stakeholders, translating telemetry, utilization, and forecast outputs into infrastructure planning recommendations and operational decision support.
- Integrated monitoring and capacity feeds from BMC TrueSight/TSCO, CA Wily, and AppDynamics into unified reporting layers used for bottleneck detection, seasonal trend analysis, and platform capacity reviews.
- Designed optimized Oracle schemas and historical retention structures for long-term telemetry history, seasonal workload patterns, capacity risk analysis, and forecast model training.
- Containerized Python ETL jobs on EC2/ECS to support repeatable on-demand processing for recurring capacity forecasting, infrastructure analytics, and reporting pipelines.
- Partnered with application, infrastructure, and operations teams to align capacity forecasts with platform demand, growth assumptions, remediation planning, and infrastructure optimization priorities.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, translating real-user and synthetic telemetry into performance and optimization recommendations.
- Led the FAST project, data-mining user performance metrics to identify opportunities adopted by engineering teams for critical revenue-generating digital systems.
- Implemented End-User Monitoring and end-to-end transaction tracing across web and mobile applications, improving operational visibility into demand, latency, and user experience patterns.
- Participated in AWS cloud migration planning by providing APM readiness assessment, monitoring requirements, and performance engineering input for cloud transition decisions.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM SME for an enterprise environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting large-scale telemetry collection and performance analysis.
- Designed dashboards, alert thresholds, SLA monitoring, and management modules that became standard patterns across client environments.
- Built Perl and Korn Shell data-extraction scripts to automate performance reporting and replace manual weekly exports.
- Provided sizing recommendations, golden-image deployment guidance, and onboarding support for new monitored applications across J2EE/WebLogic environments.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while maintaining high transaction capacity and sub-second query latency.
- Optimized C++/OCI transaction processing and database access patterns, reducing physical hardware footprint by 95% while preserving performance requirements.
- Built validation and testing frameworks to verify migration correctness across millions of transaction records during phased cutover.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt / Cairo, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, Perl, Korn Shell/Ksh, C++, Java, XML, JSON

Flagship Projects

HorizonScale — AI Capacity Forecasting Engine

Built an AI-driven capacity forecasting engine using Python, PySpark, Prophet, scikit-learn, and Streamlit to replace manual forecasting workflows, predict infrastructure bottlenecks months ahead, and support cloud-style capacity planning, utilization analysis, and executive decision support.

Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, Capacity Planning, Cloud Capacity Forecasting, Telemetry Pipelines, Utilization Analysis

AWS Serverless Lakehouse Platform

Designed a serverless AWS data platform using S3, Glue, Athena, Redshift-oriented patterns, Parquet, and PySpark to support scalable ETL, data access, and enterprise analytics workloads.

Technologies: AWS S3, AWS Glue, Athena, Amazon Redshift, PySpark, Parquet

FAST Performance Telemetry Analytics

Data-mined real-user and synthetic monitoring telemetry to produce optimization recommendations for critical digital systems and engineering stakeholders.

Technologies: Dynatrace, Gomez Synthetic Monitoring, Telemetry Analysis, Performance Engineering